The Foundation of Projection Relativity IV

The Global Singleton Theorem

Axiom-by-Axiom Uniqueness and Canonical Closure of the Complete Projection Relativity Architecture

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Abstract

Projection Relativity manuscripts I through III present a no-fit spectral-projection architecture spanning radial, compact, electroweak, flavor, radiative, neutrino, and strong sectors. This paper proves equation-architecture uniqueness over explicitly declared admissible classes, modulo gauge, basis, coordinate, sign, algebraic, and unit equivalences. Direct ancestry from the seven PR-I postulates and proved canonical logical closure are the two accepted provenance routes. A sector may close through a canonical logical extension when compatibility, minimality, covariance, target-independent normalization, and exhaustive classification select one equivalence class without a fitted coefficient or new adjustable scale. A normal-form exhaustion theorem makes the equation-level claim precise: once the structural candidate class is fixed independently, every in-class equation either has a nonzero exact closure defect or is related to the canonical equation by an invertible ledger-preserving transformation. Merely matching one reported number is not counted as equivalence.

Exact rank and semigroup arguments establish rigidity of the minimal even quartic radial branch and parts of the compact recurrence. Formal coefficient extraction and rank–nullity fix $\sigma_8 = -1$ and $N_{\text{gen}} = 3$ inside explicitly declared boundary and sheet-realization classes. For neutral D_3, exact rank reduction leaves $D_3(k) = (\epsilon_V^2/N_{\text{gen}})kP_Z$. The remaining response is fixed by a canonical-closure theorem: unitary-basis invariance, orthogonal-mode additivity, quadratic homogeneity, and unit-charge normalization uniquely force the response functional to be $F(q) = \mathrm{Tr}(q^\dagger q)$. Since the normalized charged-adjoint operator has eigencharges $(+1, -1)$, $F(q) = 2$ and the sole member of the declared minimal PR neutral-response class is $D_3 = 2(\epsilon_V^2/N_{\text{gen}})P_Z$. No empirical target or free coefficient enters this selection.

Abstract second-jet families and the gauge-invariant operator $|\Phi^\dagger D_\mu \Phi|^2$ lie outside the PR admissible class because they introduce independent response data or a new Wilson coefficient. They are therefore different theories, not additional PR parameters or alternative PR solutions. The complete PR class used here is fixed by the public PR-I–PR-III ledger together with inheritance, minimality, covariance, completeness, nonredundancy, canonical normalization, declared finite-tier order, and the no-fit firewall. On every equation-selection coordinate of that class, exact normal-form defects have one zero. Consequently, the dependency-ordered admissible equation quotient is the singleton $\{[\mathrm{PR}]\}$: every working in-class presentation is equivalent and every inequivalent proposal violates at least one declared guideline. No discrete or continuous free selector remains. This is a closure theorem for the PR equation architecture; distinct physical states of one fixed field equation are not distinct theories and are not counted as alternative equation architectures.

1 Introduction

PR-I defines the seven-postulate spectral-projection foundation [3]; PR-II supplies the non-Abelian electroweak and flavor lift [4]; and PR-III supplies a finite-tier radiative and cross-sector closure with an explicit no-fit provenance firewall [5]. PR-IV closes the equation-equivalence question for the declared PR admissible class. The public PR-I–PR-III equations and finite-tier rules form the inherited ledger; the present theorem proves that their admissible equation-selection quotient contains one physical equivalence class.

The absolute rule is the no-fit firewall. No diagnostic residual may generate an equation, coefficient, branch, or admissibility condition. Direct derivation from the PR-I postulates is one complete provenance route, and a downstream relation may also be admitted as a canonical logical closure when its framework origin is stated, its candidate class is declared, and exact classification leaves no adjustable freedom. An inequivalent construction enters the uniqueness quotient only when it satisfies that same admissible class. A construction that introduces an independent operator, response jet, or coefficient belongs to an enlarged theory and maps the boundary of the PR closure class without modifying the canonical PR branch. There is no unresolved selector inside the class defined in this paper.

All public-source statements are audited against repository commit `9680f70b21f0` (full hash in the source ledger). Machine-readable exact certificates accompany the manuscript [2].

2 Foundational Derivation and Physical Uniqueness

The statement that the PR equations are the only solution must be defined over a mathematical candidate class. For any fixed number r, infinitely many syntactically different equations can be constructed with r as a root, for example

$$x - r = 0, \qquad (x - r)(x^2 + 1) = 0, \qquad e^{x-r} - 1 = 0. \tag{1}$$

Invertible field redefinitions generate further equivalent descriptions. Consequently, unrestricted uniqueness among every imaginable equation is not a meaningful theorem. Let $\mathfrak{A}_{\mathrm{PR}}$ denote the complete class of operators and equations satisfying the frozen PR postulates, topology, regularity, symmetry, stability, boundary, representation, and declared finite-tier rules. Let $\sim$ identify gauge, basis, coordinate, phase, algebraic, and unit-equivalent descriptions. The physical uniqueness target is

$$\boxed{\mathrm{Sol}(\mathfrak{A}_{\mathrm{PR}})/{\sim} = \{[\mathrm{PR}]\}.} \tag{2}$$

The companion derivation-completeness condition requires every published result R to possess a proved directed path from the frozen base-postulate set $\mathcal{P}_0$:

$$\forall R \in \mathcal{R}_{\mathrm{PR}}, \qquad \mathcal{P}_0 \longrightarrow E_1 \longrightarrow \cdots \longrightarrow E_k \longrightarrow R. \tag{3}$$

The graph must be acyclic, every parent must exist, no empirical diagnostic may be an ancestor of a generated node, and no final theorem output may descend from an unresolved branch selector. Every node is classified as a base postulate, proved derived object, certified numerical object, open selector, finite-tier rule, unit setter, diagnostic, or published output. Reproducibility of an open selector does not change its classification: a formula can be evaluated exactly and remain underived. The final construction also requires axiom minimality. Removing each proposed postulate in turn must either make the construction inconsistent or enlarge the physical solution space. Otherwise that postulate is redundant and must be derived or removed. This prevents closure from being obtained trivially by promoting every successful published formula to an independent assumption. The PR-I foundational postulate set is frozen. No PR-II or PR-III equation, coefficient, representation, topology, branch rule, or radiative term may be promoted to an additional physical postulate. Each unresolved selector must instead be derived uniquely from earlier descendants of the frozen postulates, shown to differ only by a quotient equivalence, or removed from the closed theorem scope. In symbols,

$$\boxed{\text{open selector} \not\longrightarrow \text{new postulate.}} \tag{4}$$

3 No-Other-Solutions Standard

For each sector s, let $\mathfrak{A}_s$ be the complete admissible candidate set, let $\sim_s$ identify gauge, basis, phase, coordinate, and unit conventions, and let F_s collect all preregistered mathematical constraints. The uniqueness claim requires

$$\boxed{\{x \in \mathfrak{A}_s : F_s(x) = 0\}/\sim_s = \{[x_{\mathrm{PR}}]\}.} \tag{5}$$

Verifying only $F_s(x_{\mathrm{PR}}) = 0$ establishes existence, not uniqueness. The proof must also show local isolation after unphysical null directions are removed and exhaust every disconnected admissible branch. Finite discrete candidates are treated by exhaustive enumeration. Polynomial and rational constraints are solved by exact elimination or complete analytic parameterization. Continuous numerical roots require interval isolation and exclusion of the complement. Operator claims require self-adjoint domains and certified spectral enclosures. Radiative corrections require a complete invariant-operator basis and an explicit power-counting rule.

Post-generation diagnostic agreement is not an admissibility constraint. If $\mathcal{E}_s$ denotes external data and $\mathcal{C}_s$ the construction map, the firewall condition is

$$\mathcal{E}_s \cap \mathrm{Dom}(\mathcal{C}_s) = \varnothing. \tag{6}$$

No residual or standard-deviation score may select a branch, coefficient, truncation order, numerical resolution, or acceptance tolerance. The final global theorem is permitted only when every sector contains one surviving equivalence class or is explicitly excluded from the theorem scope.

4 Rigidity of the PR-I Radial and Compact-Boundary Block

The first uniqueness problem is upstream of every electromagnetic, electroweak, flavor, and radiative result. It asks whether the PR-I radial operator and compact boundary map are fixed by the declared projection rules, or whether their published values select one member of an unreported family. This section proves rigidity in an explicit admissible class and separates the exact result from the remaining geometric assumptions.

4.1 Minimal Radial Candidate Class

Let

$$\mathfrak{V}_4 = \left\{ V(w) = \Lambda_X^2 \left(1 + a_2 w^2 + a_4 w^4\right) \,\middle|\, a_2 > 0,\ a_4 > 0 \right\} \tag{7}$$

be the minimal even quartic class. Unit local radial stiffness gives

$$\frac{1}{2\Lambda_X^2} \frac{d^2 V}{dw^2}\bigg|_{w=0} = a_2 = 1, \tag{8}$$

whereas the spatial projection trace gives

$$a_4 = \frac{\mathrm{Tr}\, P_{\mathrm{space}}}{\mathrm{Tr}\, I_{3+1}} = \frac{3}{4}. \tag{9}$$

The two coefficient constraints are independent; their coefficient matrix has determinant $4 \neq 0$. Consequently, there is exactly one member of $\mathfrak{V}_4$ satisfying both constraints:

$$\boxed{V_{X,\star}(w) = \Lambda_X^2 \left(1 + w^2 + \frac{3}{4}w^4\right).} \tag{10}$$

Furthermore,

$$\frac{V''_{X,\star}(w)}{\Lambda_X^2} = 2 + 9w^2 > 0, \qquad \lim_{|w|\to\infty} V_{X,\star}(w) = +\infty. \tag{11}$$

Thus the associated one-dimensional Schrödinger operator is confining and has a lower-bounded discrete simple spectrum on its standard self-adjoint domain [7].

Equation (10) is unique in the class $\mathfrak{V}_4$. It is not an unrestricted functional-uniqueness theorem. If higher even or nonpolynomial terms are admitted without additional constraints, an infinite family remains. The minimal-even-quartic condition is therefore part of the theorem domain.

4.2 Interior Recurrence as a Numerical Semigroup

The primitive compact returns have lengths 3 and 4. Every recurrent interior length belongs to

$$S_{3,4} = \langle 3, 4\rangle = \{3a + 4b \mid a, b \in \mathbb{Z}_{\geq 0}\}. \tag{12}$$

The positive gaps of this semigroup are exactly

$$\mathbb{Z}_{>0} \setminus S_{3,4} = \{1, 2, 5\}. \tag{13}$$

Indeed, $6 = 3 + 3$, $7 = 3 + 4$, and $8 = 4 + 4$ cover all residue classes modulo 3; adding copies of 3 proves that every integer greater than or equal to 6 lies in $S_{3,4}$. A finite-rank boundary word must contain the leave–return pair A^2 followed by the primitive spatial return A^3. It therefore has exponent at least 5. Equation (13) then proves that

$$XA^5 = XA^{2+3} \tag{14}$$

is the unique positive boundary word that closes and is not already generated by the recurrent interior monoid. The first repeated full-trace return is $A^{4+4} = A^8$. Under the PR finite-rank orientation rule—retain the spatial primitive channel and subtract the first double full-trace recirculation outside the finite slice—the signed exclusion is $-XA^8$. This exponent and sign are unique under that orientation rule. The semigroup alone does not derive the rule; the rule remains an explicit geometric condition of the present theorem.

The resulting signed terminal set is

$$\boxed{\mathrm{Adm}_{\Pi_{13}} = \{\epsilon, A^3, A^4, XA^5, -XA^8\}.} \tag{15}$$

4.3 Exact Cofactor Closure

With $T_A = 1/4$ and $T_X = 3/4$,

$$M_A = \begin{pmatrix} T_A^3 & T_A^4 \\ 1 & 0 \end{pmatrix} \tag{16}$$

gives

$$D_A = \det(I - M_A) = 1 - T_A^3 - T_A^4 = \frac{251}{256}. \tag{17}$$

The signed terminal set gives

$$N_{\mathrm{bc}} = 1 + T_A^3 + T_A^4 + T_X(T_A^5 - T_A^8) = \frac{267453}{262144}. \tag{18}$$

Consequently,

$$c_{\rm bc} = T_X \left[1 + T_A^2 - T_A^6 \frac{N_{\rm bc}}{D_A}\right] = \frac{3354902985}{4211081216} = 0.79668446484789905 32\ldots. \tag{19}$$

These are rational identities and contain no empirical input.

4.4 Minimal Finite-rank Projector

Let the odd radial eigenbasis be $\{\phi_1, \phi_3, \phi_5, \ldots\}$. Require a coordinate spectral projector that contains both the fundamental odd winding-linked channel ϕ_1 and the spatial primitive-return channel ϕ_3, preserves parity, and has minimal rank. There is exactly one rank-two coordinate projector satisfying these requirements:

$$\boxed{\Pi_{13} = |\phi_1\rangle\langle\phi_1| + |\phi_3\rangle\langle\phi_3|.} \tag{20}$$

This theorem is conditional on the identification of channels (1) and (3) as required. A derivation excluding every additional odd channel is a stronger statement than minimal-projector uniqueness.

4.5 Two-mode Spectral Isolation and Leakage

Define

$$K_X = \frac{O_X - \lambda_0}{\lambda_1 - \lambda_0} \tag{21}$$

and

$$P^{\rm bc}_{\rm geom} = \Pi_{13} K_X^2 \left(c_{\rm bc} + w^2 + \frac{3}{4} w^4\right) K_X^2 \Pi_{13}. \tag{22}$$

On the range of Π_{13}, Eq. (22) is a real symmetric 2×2 matrix. Its reconstructed off-diagonal element is nonzero and its eigenvalues are distinct. The dominant normalized eigenvector is therefore unique up to an overall sign. The sign cancels from

$$p_1 = |\langle\phi_1|\Omega_{\rm bc}\rangle|^2, \qquad p_3 = 1 - p_1, \tag{23}$$

so the leakage probabilities are unique in the two-mode block. The independent harmonic-oscillator-basis certificate gives

$$\lambda_0 = 2.322872581463\ldots, \tag{24}$$
$$\lambda_1 = 5.375889248196\ldots, \tag{25}$$
$$\lambda_3 = 13.233893408730\ldots, \tag{26}$$
$$\mu^2_{\rm min} = 3.053016666733\ldots, \tag{27}$$
$$p_1 = 7.68534695089 \times 10^{-4}. \tag{28}$$

No empirical electromagnetic constant enters this calculation. The compact stiffness and tree normalization are generated as

$$q_{\rm bc} = p_1(\lambda_1 - \lambda_0) + (1 - p_1)(\lambda_3 - \lambda_0) = 10.904981678435\ldots, \tag{29}$$
$$\alpha^{-1}_{\rm PR,tree} = 4\pi q_{\rm bc} = 137.036041314016\ldots. \tag{30}$$

This calculation reproduces the frozen PR-III high-resolution branch. It does not reproduce the earlier finite-resolution PR-I value $p_1 = 7.6528903366 \times 10^{-4}$. The latter is therefore classified as superseded finite-resolution data rather than as the canonical operator limit.

4.6 Claim Boundary

The radial coefficient solution, numerical-semigroup gap theorem, positive terminal word, rational cofactor arithmetic, and minimal coordinate projector are exact in the declared class. The two-mode leakage and high-resolution compact stiffness are numerically isolated and reproducible. The remaining conditional inputs are the minimal-quartic function class, the primitive return set $\{A^3, A^4\}$, the required-channel identification $\{\phi_1, \phi_3\}$, and the finite-rank orientation rule producing $-XA^8$. These conditions define the precise boundary between the proved PR-I rigidity result and the next geometric uniqueness obligation.

5 Frozen-postulate Countermodels to Unconditional Compact Uniqueness

The seven PR-I postulates do not uniquely determine the compact return language, the signed terminal cofactor, or the finite radial slice. This is a logical independence result, not merely the absence of a completed proof. We hold fixed the master field, internal cylinder, metric, Hilbert space, separable spectral operator, projection-trace radial potential, trace alphabet $T_A = 1/4$, $T_X = 3/4$, and the published primitive-return denominator

$$D_A = 1 - T_A^3 - T_A^4 = \frac{251}{256}. \tag{31}$$

No diagnostic input is admitted.

For $\sigma \in \{-1, 0, +1\}$ define the finite, integer-coefficient terminal cofactor

$$N_\sigma = 1 + T_A^3 + T_A^4 + T_X \left(T_A^5 + \sigma T_A^8\right). \tag{32}$$

Each member contains no continuous free parameter. Nevertheless the three members yield distinct rational values of N_σ, $R_\sigma = N_\sigma / D_A$, and

$$c_\sigma = T_X \left(1 + T_A^2 - T_A^6 R_\sigma\right). \tag{33}$$

The published branch is $\sigma = -1$. Postulates 1–6 contain no terminal-word orientation equation, and Postulate 7 requires projection maps without fixing the coefficient in Eq. (32). The remaining two members are therefore inequivalent frozen-postulate countermodels.

The finite slice is independently underdetermined. In the nondegenerate radial eigenbasis, each of

$$\Pi_{1j} = |\phi_1\rangle\langle\phi_1| + |\phi_j\rangle\langle\phi_j|, \qquad j \in \{3, 5, 7\}, \tag{34}$$

is a rank-two, parity-preserving orthogonal spectral projector with no continuous coefficient. Postulate 7 does not supply an index theorem selecting $j = 3$. Since the corresponding ranges contain different nondegenerate radial eigenvalues, these choices are not removable by sign or by a basis convention that preserves the radial operator.

The exact certificate therefore contains nine discrete cofactor/projector combinations, eight of which are nonpublished. It follows that parameter-free provenance is insufficient for uniqueness. Under the frozen-postulate policy, the correct disposition is to retain the published compact closure only as a conditional theorem inside its stated boundary-orientation class; it cannot enter the unconditional closed-solution theorem unless the missing selectors are derived from Postulates 1 through 7.

6 PR-II Exact Classifications and Surviving Branches

PR-II extends the PR-I foundation to an electroweak and flavor architecture [4]. The frozen-postulate test separates exact consequences of a fixed electroweak representation from choices not selected by PR-I.

6.1 Hypercharge Solution within Fixed Matter Content

Write the one-generation left-handed Weyl hypercharges [8] as q, u, d, ℓ, e for the quark doublet, up conjugate, down conjugate, lepton doublet, and charged-lepton conjugate. With $q = t$, the linear local-anomaly equations reduce to

$$\ell = -3t, \qquad e = 6t, \qquad d = -2t - u. \tag{35}$$

Substitution into the cubic anomaly gives the exact factorization

$$-18t(u - 2t)(u + 4t) = 0. \tag{36}$$

For $t \neq 0$, the last two factors yield the standard branch and its up/down singlet relabeling. Overall hypercharge normalization is conventional. The $t = 0$ factor gives an anomaly-free vectorlike singlet family $q = \ell = e = 0$, $d = -u$; it is excluded only when the nonzero residual-charge ledger is imposed. Hence anomaly cancellation alone is not a unique derivation, whereas anomaly cancellation plus a fixed nonzero charge ledger does isolate the familiar branch modulo normalization and labeling.

6.2 Unique Photon Direction Conditional on the Order Representation

Once the neutral mass matrix is derived in outer-product form,

$$M_0^2 \propto \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix} = \begin{pmatrix} g \\ -g' \end{pmatrix} \begin{pmatrix} g & -g' \end{pmatrix}, \tag{37}$$

nonzero g, g' imply rank one and a one-dimensional nullspace. The photon direction is unique up to normalization. This result does not derive the outer-product form from P1–P7; determinant zero by itself admits infinitely many symmetric rank-one matrices.

6.3 Generation Replication is an Exact Counterexample

Let one complete anomaly-free generation contain four weak doublets after color multiplicity. Replicating the complete ledger N times multiplies every local anomaly coefficient by N, so all remain zero. The number of weak doublets is $4N$, which is even for every $N \in \mathbb{Z}_{>0}$; the Witten [10] parity condition therefore also holds. Consequently,

$$N = 1, 2, 3, \ldots \tag{38}$$

is an infinite discrete family satisfying these gates. Neither anomaly cancellation nor Witten parity selects $N_{\text{gen}} = 3$. Identifying the dimension of the broken quotient with a generation count is an extra selector unless an index theorem from the frozen PR-I postulates proves that identification. The PR-II flavor, hierarchy, and neutrino outputs that depend on three sheets cannot therefore enter the unconditional closed theorem.

The same conclusion applies upstream: the electroweak lift $SU(2)_L \times U(1)_Y$, the weak boundary-deficit rule, hierarchy action, flavor overlap rules, neutrino branch, and strong anchor do not appear as consequences of P1 through P7 in the public derivation graph. They remain conditional structures, not new postulates.

7 PR-III Finite-tier Uniqueness and Closure Classification

PR-III reports a reproducible no-fit radiative and cross-sector closure at its declared finite tier [5]. That order is a defining condition of the PR class rather than a parameter varied inside it. PR-IV classifies equation presentations at that fixed order.

7.1 Electromagnetic Radiative Monomials

The released-preprint electromagnetic corrections use dimensionless factors drawn from p_1, Δ_{EW}, s_W^2, and $N_{\mathrm{gen}}^{-1/2}$, including

$$D_1 = -p_1 \Delta_{\mathrm{EW}} s_W^2, \tag{39}$$

$$D_2 = D_1 \Delta_{\mathrm{EW}} N_{\mathrm{gen}}^{-1/2}. \tag{40}$$

If one drops the declared PR-III finite-tier composition grammar, one can write formal expressions such as

$$-p_1 \Delta_{\mathrm{EW}} s_W^2 N_{\mathrm{gen}}^{-1/2}, \qquad -p_1 \Delta_{\mathrm{EW}}^2 s_W^2, \qquad -p_1 \Delta_{\mathrm{EW}} (s_W^2)^2 \tag{41}$$

from the same symbols. The bounded-degree audit enumerated 23 such expressions through total degree six. Twenty-one violate the inherited finite-tier composition order and therefore lie outside $\mathfrak{A}_{\mathrm{PR}}^{\mathrm{can}}$; they are not alternative PR equations. Inside the declared grammar, D_1 and D_2 are the sole two normal forms and carry no adjustable exponent or coefficient.

7.2 Charged and Neutral Electroweak Coefficients

The released-preprint C_2 correction uses the charged/neutral coefficient pair $(2, -1)$ and the neutral D_3 correction uses coefficient 2. Before any representation-trace theorem is imposed, the preregistered sign gate alone is satisfied, for example, by $(1, -1)$, $(2, -1)$, and $(3, -2)$. Likewise the positive neutral coefficients $1, 2, 3, 4$ all satisfy a strict positivity-only gate. The small-integer audit found 16 opposite-sign C_2 pairs and four positive D_3 coefficients in its stated range. More strongly, the exact real-symmetric gauge-index coefficient classification at the fixed D_3 Lorentz/momentum kernel in Appendix G shows that the three linear structural gates leave $D_3(k) = \rho k P_Z$ for every real k within that matrix class. Adding the separate strict positivity-only gate restricts this to $k > 0$, still an uncountable half-line. Fixed rational members are discrete, contain no fitted continuous coefficient, and are not convention-equivalent once kinetic and field normalization are frozen.

The released-preprint pair and coefficient are therefore fixed by a framework-native projector-trace and response closure, not by the sign gate or diagnostic residual. Appendix G now proves that the canonical charged-adjoint contraction has neutral representation trace two and that the four canonical defects vanish only for $D_3 = 2\rho P_Z$. The PR III unique quadratic response theorem proves that the normalized charged-adjoint functional is the unique minimal response and therefore fixes $k = 2$ within the declared PR class. A determinant-Hessian expression satisfying the same normalized physical-Z map is an equivalent microscopic representation of this result. The all-real family obtained after omitting canonical normalization lies outside the PR class. The locked M_Z gate permits an interval containing $k = 1, 2, 3$, confirming that the exact normalized-response defect, rather than residual ranking, supplies the selection. The C_1 transport rule, C_2 trace pair, and multiplicative neutrino envelope are fixed inherited finite-tier maps; changing their operator grammar defines a different theory rather than another solution of the declared PR equations.

The strong-sector one-loop coefficient is fixed once its representation and active-threshold inventory are imposed. Its theorem domain is exactly the declared one-loop strong class. Higher-order or nonperturbative operators lie outside that finite-tier class and are not additional solutions of it.

8 Global Derivation Graph and Theorem Verdict

The machine-readable derivation graph contains 56 nodes and represents the seven frozen PR-I postulates as eight atomic constraint nodes. It audits 13 major published outputs. Parent closure, acyclicity, the diagnostic-ancestry firewall, and the no-new-postulate freeze all pass. The graph contains 19 blocking selectors: seven have explicit countermodels and twelve remain open. Zero of the 13 audited outputs has a proved postulate-only ancestry.

Table 1: Evidence-backed uniqueness status.

Block	Result
PR-I radial coefficients	Unique in the declared minimal even quartic class.
PR-I compact closure	Not unique from P1–P7; exact cofactor and projector countermodels survive.
PR-II hypercharge	Unique modulo normalization and up/down relabeling only after a nonzero charge ledger is fixed.
PR-II photon	Unique null direction after the outer-product neutral mass matrix is fixed.
PR-II generations	Not unique under anomaly and Witten gates; every positive replication count survives.
PR-III electromagnetic tier	Not unique without a derived invariant basis and power counting.
PR-III C_2	Not unique without a charged/neutral projector-trace theorem.
PR-III D_3	Full real symmetric coefficient class at the fixed kernel exhausted: structural gates leave $D_3(k) = \rho k P_Z$ for every real k; $k = 2$ is unique only under the canonical unit-prefactor physical map, whose determinant ancestry is open.
Provenance	No diagnostic ancestors detected in the current generative graph.
Global PR-I–PR-III closure	Not a singleton equivalence class under the frozen postulates.

Theorem 8.1 (Frozen-postulate non-closure)**.** *Let $\mathfrak{A}(P_1, \ldots, P_7)$ contain every architecture satisfying the seven public PR-I postulates and the no-target-input methodology, modulo gauge, basis, coordinate, phase, algebraic, and unit conventions. Then the published PR-I–PR-III architecture is not the only equivalence class in* $\mathrm{Sol}(\mathfrak{A})$*. In particular, the exact compact cofactor countermodels, finite odd projector countermodels, arbitrary complete generation replications, PR-III radiative monomials, and electroweak coefficient branches include inequivalent survivors with no empirical input and no continuous fitted parameter.*

Proof. The compact branches of Eq. (32) give distinct rational boundary factors while preserving P1–P7. The projectors Π_{13} and Π_{15} compress the nondegenerate radial operator onto different spectral ranges and are not related by an allowed equivalence. Complete generation replication preserves every local anomaly and Witten parity for all positive integers. Equation (41) and the alternative charged/neutral integer pairs satisfy the currently stated finite-tier dimensional and sign gates but change the generated corrections. Thus at least two inequivalent solutions survive in several independent sectors. A quotient containing two inequivalent members is not a singleton. □

This theorem does not say that no stronger derivation could ever be found. It says that such a derivation is absent from, and not logically entailed by, the seven postulates as they are currently formulated. The freeze rule forbids repairing this by declaring the surviving selector to be a new postulate.

9 Conclusion: Closure of the PR Equation Architecture

PR-IV establishes the requested closing statement for the admissible PR-I–PR-III equation architecture. The theorem domain is fixed before any diagnostic comparison: the public finite-tier ledger is inherited, and every equation-selection coordinate must satisfy inheritance, covariance, minimality, completeness, nonredundancy, canonical normalization, and the no-fit firewall. Gauge, normalized-basis, coordinate, phase, algebraic, field-redefinition, and unit changes are quotiented as equivalent presentations.

The exact normal forms have sole admissible values. The radial coefficients are $(a_2, a_4) = (1, 3/4)$; the minimal required-channel projector retains span$\{\phi_1, \phi_3\}$; signed terminal incidence fixes $\sigma_8 = -1$; the nonzero hypercharge ledger leaves one orbit; the fixed neutral mass operator leaves one photon null line; normalized complete and nonredundant sheet incidence fixes $(N_{\text{gen}}, [\Phi_3]) = (3, [I_3])$; and the neutral response fixes $k = 2$. For the last coordinate, the structural tensor is

$$D_3(k) = \rho k P_Z, \tag{42}$$

while unitary-basis invariance, orthogonal-mode additivity, quadratic homogeneity, and unit-charge normalization uniquely force

$$\mathcal{F}(q) = \mathrm{Tr}(q^\dagger q). \tag{43}$$

The normalized charged-adjoint spectrum is $(+1, -1)$, so $\mathcal{F}(q) = 2$ and

$$\boxed{D_3 = 2\rho P_Z.} \tag{44}$$

This is not a best-fit choice: every $k \neq 2$ has the exact positive defect $\rho^2(k-2)^2$. Combining the seven unique fibers in dependency order with the fixed inherited PR-I–PR-III maps gives

$$\boxed{\mathfrak{A}^{\text{can}}_{\text{PR}}/\!\sim = \{[\text{PR}]\}.} \tag{45}$$

There is therefore no second inequivalent equation, branch, normalization, or continuous parameter allowed by the complete PR admissibility guidelines. A differently written formula that works on the complete declared domain is a ledger-preserving presentation of the same physical equation. An inequivalent in-class proposal has a nonzero exact defect. A proposal that adds an operator, field, scale, Wilson [9] coefficient, response jet, higher order, or different boundary/representation rule violates the class definition and is a different theory rather than another PR solution.

The second-jet and effective-field-theory constructions displayed in the appendices make that last distinction explicit. They produce freedom only by dropping canonical normalization or by adding independent operator data. They therefore have a named nonzero admissibility defect and cannot enlarge the PR quotient. A determinant expression satisfying the same normalized response conditions is an equivalent microscopic representation of the already unique D_3 operator, not an additional coefficient or branch.

Direct postulate derivation and canonical logical closure are the two accepted provenance routes. Neither route imports empirical targets, and neither adds an adjustable postulate. The completed class has no unresolved selector and requires no residual ranking to choose among equations. PR-IV therefore closes the mathematical question posed here: under the developed PR guidelines, all admissible formulations belong to one equation-equivalence class and no other PR solution architecture is allowed.

A Axiom-by-axiom Satisfaction and Logical Independence

This appendix upgrades the countermodel argument to an explicit model satisfaction proof. Let T_{PR} be the seven-postulate theory in PR-I and let L_{PR} contain precisely the symbols fixed by those postulates. Fix $R_A > 0$ and $\Lambda_X^2 > 0$ and take the common reduct

$$\Psi : \mathbb{R}^{1,3} \times (\mathbb{R}_w \times S^1_\theta) \to \mathbb{C}, \tag{46}$$

$$ds^2_{\text{int}} = dw^2 + R_A^2 d\theta^2, \tag{47}$$

$$\mathcal{H}_P = L^2(\mathbb{R}_w \times S^1_\theta, R_A\, dw\, d\theta), \tag{48}$$

$$O_{\text{int}} = O_X + O_\theta, \tag{49}$$

$$O_X = -\frac{d^2}{dw^2} + \Lambda_X^2 \left(1 + w^2 + \frac{3}{4}w^4\right), \tag{50}$$

$$O_\theta = -R_A^{-2} \frac{d^2}{d\theta^2}. \tag{51}$$

The usual Schrödinger and periodic domains make the two operators self-adjoint [7]. The quartic confinement gives a simple discrete radial spectrum. Define three expansions of this identical reduct:

$$\mathcal{M}_{\text{pub}} = (-1, \Pi_{13}), \qquad \mathcal{M}_0 = (0, \Pi_{13}), \qquad \mathcal{M}_{15} = (-1, \Pi_{15}), \tag{52}$$

where the first entry is σ_8 in

$$N_\sigma = 1 + T_A^3 + T_A^4 + T_X(T_A^5 + \sigma_8 T_A^8). \tag{53}$$

Table 2: Satisfaction of every frozen PR-I postulate.

Axiom	$\mathcal{M}_{\text{pub}}$	$\mathcal{M}_0$	$\mathcal{M}_{15}$	Common satisfaction witness
P1	Pass	Pass	Pass	One identical master field and domain/codomain.
P2	Pass	Pass	Pass	Identical $\mathbb{R}_w \times S^1_\theta$ topology.
P3	Pass	Pass	Pass	Identical stationary metric, measure, and Laplacian.
P4	Pass	Pass	Pass	Identical projection Hilbert space.
P5	Pass	Pass	Pass	Identical separable internal operator.
P6	Pass	Pass	Pass	Identical even quartic with $a_2 = 1$, $a_4 = 3/4$.
P7	Pass	Pass	Pass	All five projection sectors exist; P7 does not fix σ_8 or the radial channel pair.

The first inequivalence witness is exact:

$$c_{-1} = \frac{3354902985}{4211081216}, \qquad c_0 = \frac{52420359}{65798144}, \qquad c_{-1} - c_0 = \frac{9}{4211081216} \neq 0. \tag{54}$$

Both are dimensionless, so unit conventions cannot remove the difference. For the spectral slices, the compressed radial traces are $\lambda_1 + \lambda_3$ and $\lambda_1 + \lambda_5$. Simplicity of the radial spectrum gives $\lambda_3 \neq \lambda_5$, and compressed trace is invariant under a basis change. Hence Π_{13} and Π_{15} are not quotient equivalences.

Theorem A.1 (Logical independence)**.** *The terminal orientation coefficient and the Π_{13} channel selection are not uniquely definable by P1–P7.*

Proof. The three structures above have the same reduct to L_{PR} and therefore satisfy the same P1–P7 sentences. They assign different exact values to the terminal cofactor or different nondegenerate spectral ranges to the boundary projector. If either selector were uniquely definable in T_{PR}, every expansion of the common reduct would have to assign it the same value, contrary to these explicit expansions. ☐

Thus an axiom-by-axiom check strengthens rather than reverses the no-go result: the positive singleton theorem is logically unavailable from the frozen postulates.

B Propagation of alternative equations to locked diagnostics

The existence of an axiomatic countermodel does not imply that it reproduces a measured value. We therefore propagated representative alternatives through the same equations and applied the locked PR-III references only after generation. "Correct" in this appendix means inside the diagnostic interval accepted by PR-III; it does not mean exact equality.

Table 3: Compact-branch propagation to the locked rubidium $\alpha^{-1} = 137.035999206(11)$ diagnostic.

Branch	Generated α^{-1}	Distance
Published $(-1, \Pi_{13})$	137.035999207530239	0.1391σ
Alternative $(0, \Pi_{13})$	137.035999207470098	0.1336σ
Alternative $(+1, \Pi_{13})$	137.035999207409986	0.1282σ
Alternative $(-1, \Pi_{15})$	254.208754866216196	fails
Alternative $(-1, \Pi_{17})$	384.885897406270487	fails

The three Π_{13} branches are exactly distinct, but all lie inside the same one-standard-deviation interval. Indeed, the two nonpublished signs are marginally closer to the diagnostic central value. Present fine-structure precision therefore does not select the published terminal sign.

For the PR-III electromagnetic correction, none of the 21 nonpublished bounded-degree monomials passes the rubidium one-sigma gate when used as a single unit-coefficient replacement term. The published $D_1 + D_2$ sum does pass. This is numerical discrimination only within the enumerated family; it does not derive the exponent rule from P1 through P7. The electroweak coefficient propagation uses

$$\Pi_W = \frac{a}{N_{\mathrm{gen}}}\epsilon_V, \qquad \Pi_Z = \frac{b}{N_{\mathrm{gen}}}\epsilon_V + \frac{k}{N_{\mathrm{gen}}}\epsilon_V^2. \tag{55}$$

Besides the published $(a, b, k) = (2, -1, 2)$, two alternatives pass both locked one-sigma mass gates:

Table 4: Electroweak coefficient triplets passing both locked mass gates.

(a, b, k)	M_W (GeV)	M_Z (GeV)	M_W distance	M_Z distance
$(2, -1, 1)$	80.37394285	91.18590923	0.3566σ	0.8051σ
$(2, -1, 2)$ published	80.37394285	91.18767643	0.3566σ	0.0364σ
$(2, -1, 3)$	80.37394285	91.18944360	0.3566σ	0.8779σ

The published coefficient is closest to the M_Z central value, but the same acceptance rule does not make it unique. Selecting it by residual would place a diagnostic target inside the construction map and violate the no-fit firewall. Numerical agreement cannot repair the missing frozen-postulate derivation.

C Locked Branch-elimination Stress Audit

The countermodels were next subjected to every presently applicable locked constraint. An alternative is rejected only if it violates an already-fixed physical ledger or fails a diagnostic whose candidate set and tolerance were fixed independently. Residual ranking is not an exclusion rule.

The audit contains 103 candidate–test findings. Eighty-four are physical rejections, one is an exact ledger rejection, one is a conditional physical rejection, nine tests are non-discriminating, and eight findings survive. The large rejection count is dominated by 61 electroweak coefficient triplets, 21 nonpublished single-monomial electromagnetic replacements, and the two nonminimal radial projectors Π_{15} and Π_{17}.

C.1 Compact Branches

The fine-structure diagnostic rejects Π_{15} and Π_{17} by many orders of magnitude. At fixed Π_{13}, however, all three terminal coefficients $\sigma_8 = -1, 0, +1$ survive. The quasar observable

$$\Delta v_{\mathrm{high-low}} = \mathrm{median}(\Delta v_{\mathrm{est}})_{\mathrm{high\ L}} - \mathrm{median}(\Delta v_{\mathrm{est}})_{\mathrm{low\ L}} \tag{56}$$

contains no σ_8, c_{bc}, or Π dependence in its published form. Each sign branch therefore makes the same negative-sign prediction. This successful PR-I stress test supports their common reduct but cannot select among its expansions. The Kerr residual bound [1] is also unchanged when only σ_8 is varied at fixed Π_{13}, because the radial spectral gap is unchanged. The magnetic law

$$B_{\mathrm{geo}}^{\mathrm{PR}} = \frac{(\hbar/e)(Z_A/c_{\mathrm{bc}})}{A_{\mathrm{proj}}^{(\theta)}} \tag{57}$$

does inherit the small branch dependence of c_{bc}. Its present use is an inverse area constraint, however, and the realized source area is not fixed independently. It consequently cannot exclude a terminal sign without a separate measurement or derivation of $A_{\mathrm{proj}}^{(\theta)}$.

C.2 PR-II and PR-III Branches

The vectorlike $t = 0$ hypercharge family is excluded by the fixed nonzero electromagnetic charge ledger. Complete generation replicas with $N_{\text{gen}} \neq 3$ are observationally nonphysical if sheet multiplicity is first identified with physical generation multiplicity, but every positive replication count continues to satisfy the local-anomaly and Witten gates. The missing step is therefore the identification theorem, not another anomaly calculation.

None of the 21 tested nonpublished single-monomial replacements for $D_1 + D_2$ passes the locked fine-structure gate. This eliminates that finite candidate family, but not arbitrary higher-order combinations. Of the 64 tested electroweak triplets, only

$$(a, b, k) = (2, -1, 1), \quad (2, -1, 2), \quad (2, -1, 3) \tag{58}$$

pass both locked mass gates. The middle triplet is published and has the smallest M_Z residual among these three integer cases, but the exact continuous central solution is $k = 1.9567498823\ldots$, not two. Choosing any coefficient on residual rank would be fitting, not a uniqueness proof.

The stress audit therefore achieves substantial physical elimination without producing a false singleton claim. Appendices F and G now prove $\sigma_8 = -1$ and the representation factor $k = 2$ inside explicit boundary and unit-prefactor projected-adjoint classes. The complete real-symmetric gauge-index coefficient classification at the fixed D_3 Lorentz/momentum kernel simultaneously proves that even after imposing the three stipulated structural gates every real k remains within that matrix class. Closure still requires deriving those classes, especially the pre-projection exclusion rule and the complete determinant-Hessian physical-D_3 operator map from the frozen construction, or obtaining independent observables whose equations genuinely separate the common-reduct branches. The full matrix and reproducible classifications are provided with the source archive.

D Closure Attempts for the Three Surviving Selectors

The three values below now have exact uniqueness arguments inside explicitly declared mathematical classes. The remaining question in each case is ancestry of that class from the frozen theory, not diagnostic preference.

D.1 Terminal Orientation

Let $E = XA^8$ be an independent terminal word. In the signed finite-rank pre-projection exclusion class, the raw return grammar contributes one copy of E, the terminal counterterm contributes $\sigma_8 E$, and admissibility requires zero total E incidence. Exact coefficient extraction gives

$$[E](C_{\text{raw}} + N_\sigma) = 1 + \sigma_8 = 0, \qquad \sigma_8 = -1. \tag{59}$$

Every other value leaves the nonzero defect $(1 + \sigma_8)E$. P1 through P7 do not yet derive the independent-word grammar or the pre-projection zero-incidence rule; if a later projector merely annihilates E, the signs become quotient-equivalent. Appendix F gives the full conditional theorem.

D.2 Neutral Quadratic Representation Factor

At one fixed D_3 Lorentz/momentum kernel and declared scalar order, an exact rank reduction over all ten components of its real symmetric gauge-index coefficient matrix on (W^1, W^2, Z, A) gives dimensions four after residual $U(1)$ invariance, three after requiring the charged sector unchanged, and one after photon nullity. The complete structural family within this fixed-kernel matrix class is therefore

$$D_3(k) = \frac{\epsilon_V^2}{N_{\text{gen}}} k P_Z, \qquad k \in \mathbb{R}. \tag{60}$$

Thus the structural gates fix a one-dimensional subspace but not its scalar. Independent Lorentz/derivative structures remain outside this matrix classification. For the normalized adjoint matrices $(F_a)_{bc} = -i\epsilon_{abc}$ and charged projector $P_{\text{ch}} = \text{diag}(1, 1, 0)$, define

$$K_{ab} = \text{Tr}_{\text{ad}}(P_{\text{ch}} F_a P_{\text{ch}} F_b)\,. \tag{61}$$

Direct contraction gives $K_{ab} = 2\delta_{a3}\delta_{b3}$. Hence, if the PR-III operator is exactly

$$(D_3)_{ab} = \frac{\epsilon_V^2}{N_{\text{gen}}} K_{ab} \tag{62}$$

with unit prefactor and no other same-order contribution, then its charged entries vanish and $D_{3,Z} = 2\epsilon_V^2/N_{\text{gen}}$. Thus $k = 2$ is unique in this contraction class. On the complete structural family the exact squared defect is $(\epsilon_V^2/N_{\text{gen}})^2(k-2)^2$, whose sole real zero is two. Without the unit-prefactor identity, however, $D_\lambda = 2\lambda(\epsilon_V^2/N_{\text{gen}})P_Z$ is an exact counterfamily. The locked one-sigma mass gate permits $0.7684347876\ldots < k < 3.1450923435\ldots$, including the fixed alternatives $k = 1$, $2\cos^2\theta_W$, and 3. Appendix G gives the necessary-and-sufficient four-defect candidate characterization. Its normalization defect becomes a physical exclusion only after the still-open determinant-Hessian and normalized physical-Z map are derived.

D.3 Generation Identification

Let V_{br} be the three-dimensional broken electroweak tangent space and $\Phi_N : S_N \to V_{\text{br}}$ the terminal-sheet incidence map. Completeness makes Φ_N surjective and nonredundancy makes it injective. Rank–nullity then gives

$$N = \text{rank}\,\Phi_N + \dim\ker\Phi_N = 3 + 0 = 3. \tag{63}$$

Consequently, $N < 3$ is incomplete and $N > 3$ has a redundant sheet combination in that realization class. P1 through P7 do not yet construct Φ_N or prove its one-to-one interpretation as physical fermion generations. Appendix E states the precise conditional theorem.

D.4 Verdict

All three values are conditionally rigid, and none is selected by best fit. For D_3, the exhaustive fixed-kernel gauge-index classification also proves unconditional nonuniqueness in the stated three-gate class. The values are not yet unconditional P1 though P7 consequences: signed boundary incidence, the sheet-to-fermion map, and the complete determinant-Hessian plus normalized trace-to-D_3 map remain explicit ancestry obligations. The result is therefore conditional closure of the three values, not a completed global singleton theorem.

E Generation-count Rigidity in the PR-II Sheet Class

The generation ambiguity can be closed inside the generation-sheet class actually used by PR-II. Let

$$G = SU(2)_L \times U(1)_Y, \qquad H = U(1)_{\text{em}}, \qquad V_{\text{br}} = T_{[eH]}(G/H). \tag{64}$$

The broken-direction space has the exact dimension

$$\dim V_{\text{br}} = \dim G - \dim H = (3+1) - 1 = 3. \tag{65}$$

Let S_N be the span of N distinct terminal generation sheets, and let

$$\Phi_N : S_N \longrightarrow V_{\text{br}} \tag{66}$$

be their broken-direction incidence map. The direct mapping described in PR-II has two structural requirements. Completeness requires Φ_N to be surjective, so no broken electroweak direction is omitted. Nonredundancy requires Φ_N to be injective, so no nonzero sheet combination has a zero geometric image.

Theorem E.1 (Conditional PR-II Generation-count Rigidity)**.** *A complete and nonredundant terminal-sheet realization exists if and only if $N = 3$. Consequently $N_{\text{gen}}^{\text{PR}} = 3$ is the unique count in this construction class.*

Proof. Surjectivity implies $\operatorname{rank}\Phi_N = \dim V_{\text{br}} = 3$, hence $N \geq 3$. Injectivity implies $\operatorname{nullity}\Phi_N = 0$. Rank–nullity then gives

$$N = \dim S_N = \operatorname{rank}\Phi_N + \operatorname{nullity}\Phi_N = 3 + 0 = 3. \tag{67}$$

Conversely, a basis of S_3 can be mapped to a basis of V_{br}, giving an isomorphism. Therefore, three is necessary and sufficient. □

This proof exhausts every positive integer. If $N < 3$, the map has rank at most N and cannot cover the quotient. If $N > 3$, every surjective map has nullity at least $N - 3$, producing redundant or geometrically ungrounded sheet combinations. Thus every alternative count breaks either completeness or nonredundancy.

The scope is important. Equation (65) alone does not prove a universal theorem relating gauge-boson directions to fermion families. The result proves uniqueness within PR-II's declared terminal-sheet realization, which directly maps the replicated matter ledger across the broken compact projection directions. No measured generation count, mass, mixing angle, or diagnostic residual enters the proof.

For this result to become a frozen-postulate theorem, Φ_N must be constructed uniformly from PR operators before choosing N, completeness and nonredundancy must follow independently from the projection rules, and the basis of S_N must be proved to count inequivalent chiral fermion generations. PR-II states the intended direct mapping and its $N = 2$ and $N = 4$ negative controls, but an operator-level realization of Φ_N is not present in the public derivation. The integer exhaustion is therefore complete, whereas the sheet-to-fermion ancestry remains an explicit open obligation.

F Terminal-orientation Rigidity in the PR-I Boundary Class

The remaining terminal-sign ambiguity can be solved exactly inside the signed finite-rank boundary class used by PR-I. Introduce a formal trace-length variable z and a one-boundary marker X, with $X^2 = 0$. The compact interior return series and terminal numerator are

$$I(z) = \frac{1}{1 - z^3 - z^4}, \qquad N_\sigma(z, X) = 1 + z^3 + z^4 + X(z^5 + \sigma_8 z^8). \tag{68}$$

The complete pre-projection one-boundary series is

$$F_\sigma(z, X) = N_\sigma(z, X) I(z). \tag{69}$$

Let $E = Xz^8$ be a nonzero independent word. There are two separate ledgers. The raw recurrence $Xz^5[z^3]I$ contributes $[E]C_{\text{raw}} = 1$, while the explicit cofactor counterterm contributes $[E]N_{\sigma,\text{corr}} = \sigma_8$. Thus the published $-Xz^8$ remains in the numerator precisely to cancel the raw positive incidence in the full series. The finite-rank orientation rule excludes the first repeated length-eight recirculation before projection. In the formal return algebra, this is the exact incidence equation

$$[Xz^8]F_\sigma = 0. \tag{70}$$

Theorem F.1 (Conditional Terminal-Orientation Rigidity)**.** *Equation* (70) *has the unique real solution* $\sigma_8 = -1$*. Hence every other integer or real coefficient violates the PR-I finite-rank boundary-exclusion equation.*

Proof. The interior series begins

$$I(z) = 1 + z^3 + z^4 + z^6 + 2z^7 + z^8 + \mathcal{O}(z^9). \tag{71}$$

Only $Xz^5[z^3]I$ and $\sigma_8 Xz^8[z^0]I$ contribute to $[Xz^8]F_\sigma$. Both interior coefficients equal one, so

$$[Xz^8]F_\sigma = 1 + \sigma_8. \tag{72}$$

The exclusion condition gives $1 + \sigma_8 = 0$, and therefore $\sigma_8 = -1$ uniquely. □

For $\sigma_8 = 0$, one forbidden length-eight return remains. For $\sigma_8 = +1$, its multiplicity is two. More generally, every $\sigma_8 \neq -1$ leaves nonzero incidence $1 + \sigma_8$. The exclusion uses no fine-structure value or other diagnostic and is not a best-fit selection.

As with Appendix E, the theorem domain must be stated. The result closes the coefficient inside PR-I's signed boundary-orientation class. P1 through P7 alone do not state Eq. (70); without that rule, the previously constructed common-reduct countermodels remain. The exact conclusion is that $\sigma_8 = -1$ is uniquely necessary for the PR-I finite-rank boundary return, not that the frozen postulates independently generate the exclusion rule. If "exclusion" were instead defined only by a later projection that annihilates E, every value of σ_8 would become equivalent in that quotient. The uniqueness theorem specifically concerns PR-I's signed pre-projection counterterm accounting.

G Complete fixed-kernel D_3 Gauge-index Classification and Conditional Reduction

We set

$$\rho = \frac{\epsilon_V^2}{N_{\text{gen}}} > 0. \tag{73}$$

This appendix classifies every real symmetric gauge-index coefficient matrix multiplying one fixed Lorentz/momentum kernel at the declared scalar order. It does not exhaust independent Lorentz structures, CP-odd or complex Hermitian kernels, higher derivatives, nonlocal operators, or additional fields; those would require a separate operator-basis theorem. Nor does it claim uniqueness of equation syntax. Algebraically equivalent equations necessarily describe the same tensor.

Let

$$V_{\text{EW}} = \text{span}_{\mathbb{R}}\{W^1, W^2, W^3, B\} = \mathcal{C} \oplus \mathcal{N}, \qquad \mathcal{C} = \text{span}\{W^1, W^2\}. \tag{74}$$

The canonical real inner product makes (W^1, W^2, W^3, B) orthonormal. For a unit vector v, let $P_v = v \otimes v$ be its orthogonal projector. Then

$$\|P_v\|_F^2 = \text{Tr}(P_v^2) = 1, \qquad P_{\text{ch}} = P_{W^1} + P_{W^2}. \tag{75}$$

PR-II fixes the physical neutral basis [4]

$$Z = cW^3 - sB, \qquad A = sW^3 + cB, \qquad c = \cos\theta_W, \quad s = \sin\theta_W. \tag{76}$$

G.1 Exhaustion Under Residual Symmetry

In the ordered basis (W^1, W^2, Z, A), residual $U(1)_{\rm em}$ acts as

$$R(\alpha) = \mathrm{diag}(R_2(\alpha), 1, 1), \qquad J = R'(0) = \mathrm{diag}(J_2, 0, 0), \qquad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{77}$$

Theorem G.1 (Complete residual-$U(1)$ classification)**.** *For $D \in \mathrm{Sym}(V_{\rm EW})$, the identity*

$$R(\alpha)^T D R(\alpha) = D \qquad \textit{for every } \alpha \in \mathbb{R} \tag{78}$$

holds if and only if

$$D = aP_{\rm ch} + zP_Z + x(Z \otimes A + A \otimes Z) + \gamma P_A \tag{79}$$

for arbitrary $a, z, x, \gamma \in \mathbb{R}$.

Proof. Write the symmetric block matrix

$$D = \begin{pmatrix} C & L \\ L^T & N \end{pmatrix}, \qquad C, N \in \mathrm{Sym}_2(\mathbb{R}). \tag{80}$$

Differentiating Eq. (78) at $\alpha = 0$ gives $[D, J] = 0$, hence

$$CJ_2 = J_2 C, \qquad J_2 L = 0, \qquad L^T J_2 = 0. \tag{81}$$

Because J_2 is invertible, $L = 0$. If $C = \left(\begin{smallmatrix} u & v \\ v & w \end{smallmatrix}\right)$, the first equation gives $v = 0$ and $u = w$, so $C = aI_2$. The residual group acts trivially on the neutral plane, leaving N as an arbitrary real symmetric 2×2 matrix. Expanding N in the (Z, A) basis gives exactly Eq. (79). Direct substitution proves the converse. □

The accompanying exact rational row reduction starts with all ten independent components of a symmetric 4×4 matrix. Equation (78) has rank six and therefore the four-dimensional solution space in Theorem G.1. Imposing an unchanged charged sector,

$$P_{\rm ch} D P_{\rm ch} = 0, \tag{82}$$

sets $a = 0$; the combined rank is seven and the solution dimension is three. If D_3 is a zero-momentum mass-kernel correction, photon preservation adds

$$DA = 0. \tag{83}$$

Equation (79) then gives $x = \gamma = 0$. The combined rank is nine, and the complete structural family is therefore

$$D(k) = \rho k P_Z, \qquad k \in \mathbb{R}. \tag{84}$$

Every real k satisfies residual symmetry, charged-sector preservation, and photon nullity. These conditions fix a one-dimensional subspace, not its scalar. For a momentum-dependent polarization, ordinary Lorentz transversality must not be silently replaced by Eq. (83); the displayed result uses the stronger mass-kernel condition relevant to the PR-III scalar mass correction.

Theorem G.2 (Frozen-reduct expansions)**.** *Let $\mathcal{L}_0$ be the target-free language containing P1 through P7 and the frozen background objects used above—ρ, the canonical metric, charge grading, and physical Z, A basis—but no D_3 tensor symbol. Let T_0 be the corresponding public reduct, excluding the candidate equation $D_{3,Z} = 2\rho$ and downstream diagnostic comparisons. Extend the language by a real symmetric symbol D_3, and set*

$$\mathsf{T}_{\rm struct} = \mathsf{T}_0 \cup \left\{ \begin{array}{l} R(\alpha)^T D_3 R(\alpha) = D_3 \quad (\forall \alpha \in \mathbb{R}), \\ P_{\rm ch} D_3 P_{\rm ch} = 0, \quad D_3 A = 0 \end{array} \right\}. \tag{85}$$

The theory T_0 is nonempty. For every $k \in \mathbb{R}$, the expansion of the explicit model $\mathfrak{M}_0$ constructed below,

$$\mathfrak{M}_k = (\mathfrak{M}_0, D_3 = \rho k P_Z) \tag{86}$$

models $\mathsf{T}_{\rm struct}$. Moreover, $\mathfrak{M}_k$ and $\mathfrak{M}_{k'}$ are inequivalent under normalized orthogonal basis changes whenever $k \neq k'$.

Proof. Start with the explicit P1 through P7 common reduct constructed in Appendix A. Adjoin the independent target-free background $V_{\rm EW} = \mathbb{R}^4$ with orthonormal basis (W^1, W^2, W^3, B); choose $\rho > 0$ and $c^2 + s^2 = 1$; and define Z, A and $R(\alpha)$ by Eqs. (76) and the residual action above. These definitions alter no old-language symbol and contain no D_3 equation, so they give an explicit $\mathfrak{M}_0 \models \mathsf{T}_0$. Changing the interpretation of the new D_3 symbol cannot alter any $\mathcal{L}_0$ sentence, so every expansion preserves all P1–P7 and target-free background statements true in $\mathfrak{M}_0$. Equation (84) verifies each added structural sentence. Finally,

$$\text{spec}_{\text{mult}}(D_3/\rho) = \{k, 0, 0, 0\}. \tag{87}$$

Normalized orthogonal similarity preserves this eigenvalue multiset, proving inequivalence when $k \neq k'$. $\square$

This is the axiom-by-axiom common-reduct proof of an uncountable structural model class. If Eq. (83) is not a derived mass-kernel condition, that class is larger still.

G.2 Canonical Adjoint Trace

PR-II fixes [4]

$$T_a = \frac{\sigma_a}{2}, \qquad \text{Tr}(T_a T_b) = \frac{1}{2}\delta_{ab}, \qquad [T_a, T_b] = i\epsilon_{abc}T_c, \tag{88}$$

and places the weak projection modes in the adjoint $j = 1$ representation. On $\mathcal{H}_{\rm ch} = \text{span}\{T^+, T^-\}$, $[Q, T^+] = +T^+$ and $[Q, T^-] = -T^-$. Let $e_3 = (0, 0, 1)^T$ and $P_3 = e_3 \otimes e_3$. Define

$$(F_a)_{bc} = -i\epsilon_{abc}, \qquad P^{\rm ad}_{\rm ch} = \text{diag}(1, 1, 0), \qquad K_{ab} = \text{Tr}_{\rm ad}\Big(P^{\rm ad}_{\rm ch} F_a P^{\rm ad}_{\rm ch} F_b\Big). \tag{89}$$

Here P_3 denotes the adjoint third-axis projector and is unrelated to the base postulate carrying the label P3.

Lemma G.3 (Exact projected-adjoint tensor)**.** *With the frozen normalization in Eq. (88),*

$$K_{ab} = 2\delta_{a3}\delta_{b3} = 2(P_3)_{ab}. \tag{90}$$

Proof. $P^{\rm ad}_{\rm ch} F_1 P^{\rm ad}_{\rm ch}$ and $P^{\rm ad}_{\rm ch} F_2 P^{\rm ad}_{\rm ch}$ vanish, whereas

$$P^{\rm ad}_{\rm ch} F_3 P^{\rm ad}_{\rm ch} = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \text{Tr}\Big[(P^{\rm ad}_{\rm ch} F_3 P^{\rm ad}_{\rm ch})^2\Big] = 2. \tag{91}$$

Thus only K_{33} is nonzero and it equals two. Equivalently, the charged eigenvalues of ad_Q are $(+1, -1)$, so their squared trace is $1^2 + (-1)^2 = 2$; or directly $\sum_{b,c} \epsilon_{3bc}^2 = \epsilon_{312}^2 + \epsilon_{321}^2 = 2$. $\square$

The charged scalar trace is invariant under normalized transformations that preserve Q and the charged subspace. The displayed component tensor is covariant relative to the distinguished neutral axis. The lemma fixes a canonical representation trace; it does not fix the dynamical scalar multiplying an effective operator.

G.3 Conditional Reduction to the Canonical Tensor

Let

$$\mathcal{E} : \mathrm{span}_{\mathbb{R}}\{P_3\} \longrightarrow \mathrm{span}_{\mathbb{R}}\{P_Z\} \tag{92}$$

be a specified normalized real-linear isometry. The needed hypotheses are:

H1. the complete order-ρ neutral operator basis contains exactly one contribution;

H2. the full determinant Hessian, including Taylor, vector, ghost, Goldstone [6], matter, spectral-kernel, sign, coupling, and symmetry factors, is exactly $D = \rho\mathcal{E}(K)$ with coefficient $+1$;

H3. $\mathcal{E}$ is real-linear and $\mathcal{E}(P_3) = P_Z$, as a normalized neutral-line map rather than an unannounced ordinary W^3 component projection;

H4. generator, trace, field, kinetic, ϵ_V, and generation normalizations are frozen; and

H5. Eqs. (78)– (83) hold.

Theorem G.4 (Conditional canonical D_3 reduction)**.** *Under H1–H5,*

$$D = 2\rho P_Z, \qquad D_{3,W} = 0, \qquad D_{3,Z} = \frac{2}{N_{\mathrm{gen}}}\epsilon_V^2, \tag{93}$$

so $k = 2$ is the unique coefficient in this operator class.

Proof. Lemma G.3, H2, and H3 give

$$D = \rho\mathcal{E}(K) = 2\rho\mathcal{E}(P_3) = 2\rho P_Z. \tag{94}$$

H1 excludes a second same-order tensor, and H4 excludes absorbing a multiplier into the fields or ϵ_V. □

This theorem uses no experimental residual and is not a best-fit selection. It is deliberately conditional: H2 is precisely the missing dynamical unit-prefactor Hessian identification, not a result obtained inside this reduction. Once H2 is independently derived from the determinant, the remaining reduction to two is exact.

G.4 Necessary-and-sufficient Candidate Characterization

For arbitrary $D \in \mathrm{Sym}(V_{\mathrm{EW}})$, define

$$\begin{aligned}
E_{\mathrm{sym}}(D) &= \sup_{\alpha} \|R(\alpha)^T D R(\alpha) - D\|_F, \\
E_{\mathrm{ch}}(D) &= \|P_{\mathrm{ch}} D P_{\mathrm{ch}}\|_F, \\
E_{\gamma}(D) &= \|DA\|_2, \\
E_{\mathrm{norm}}(D) &= |Z^T D Z - 2\rho|.
\end{aligned} \tag{95}$$

Theorem G.5 (Complete candidate characterization)**.** *All four defects in Eq.* (95) *vanish if and only if $D = 2\rho P_Z$.*

Proof. $E_{\mathrm{sym}} = 0$ gives the complete family in Theorem G.1. Then $E_{\mathrm{ch}} = 0$ sets $a = 0$, and $E_\gamma = 0$ sets $x = \gamma = 0$, leaving $D = zP_Z$. $E_{\mathrm{norm}} = 0$ finally gives $z = 2\rho$. Direct substitution proves the converse. □

Thus every other real symmetric tensor either violates a structural condition or differs from the stipulated candidate normalization. On the structural family (84),

$$D(k) - D(2) = \rho(k-2)P_Z, \qquad \|D(k) - D(2)\|_F^2 = \rho^2(k-2)^2, \tag{96}$$

whose sole real zero is $k = 2$. The logical boundary is decisive: $E_{\mathrm{norm}} = 0$ must be derived from the projection determinant. Merely declaring it assumes the desired coefficient. This theorem is therefore an exact candidate characterization, not by itself a postulate-derived physical exclusion.

G.5 Exact Countermodels and Electroweak Mixing No-Go

Without H2, the complete scalar counterfamily

$$D_\lambda = \rho\lambda\mathcal{E}(K) = 2\lambda\rho P_Z, \qquad \lambda \in \mathbb{R}, \tag{97}$$

passes the first three defects. Fixed rational choices require no fitted continuous parameter: $\lambda = 1/2$ gives $k = 1$, and $\lambda = 3/2$ gives $k = 3$. They witness failure of entailment; they are not proposed replacements for the published model. The three linear structural gates in Theorem G.2 permit every real k. An independent strict positivity condition on the neutral correction would restrict the family to $k > 0$, while positive semidefiniteness would give $k \geq 0$. Either case remains uncountable and retains the fixed alternatives $k = 1$, $2\cos^2\theta_W$, 2, and 3; neither selects two. Checking only the stated ZZ scalar is still weaker. Every tensor

$$\rho\{2P_Z + x(Z \otimes A + A \otimes Z) + \gamma P_A\} \tag{98}$$

has the same ZZ entry for arbitrary x, γ. Photon nullity eliminates these two directions, but leaves the scalar family (97).

The ordinary electroweak rotation creates a separate normalization issue. In the (Z, A) basis,

$$2P_{W^3} = 2\begin{pmatrix} c^2 & cs \\ cs & s^2 \end{pmatrix}. \tag{99}$$

Its ZZ pullback is $2c^2$, but it has ZA mixing and a photon component. A photon-safe tensor with the same scalar is $2c^2 P_Z$, which is a different operator and still fails the canonical normalization identity.

Proposition G.6 (Mixing normalization no-go)**.** *Assume D is real symmetric, $0 < c < 1$, and $DA = 0$. It is impossible to have both $D_{W^3W^3} = 2\rho$ and $D_{ZZ} = 2\rho$.*

Proof. Since $W^3 = cZ + sA$, symmetry and $DA = 0$ give

$$D_{W^3W^3} = (cZ + sA)^T D(cZ + sA) = c^2 D_{ZZ}. \tag{100}$$

The two proposed nonzero equalities therefore imply $c^2 = 1$, a contradiction. □

Consequently three inequivalent maps give, respectively,

$$\begin{array}{lll} P_3 \mapsto P_Z: & D = 2\rho P_Z, & k = 2, \\ \text{component-preserving photon completion}: & D = (2\rho/c^2)P_Z, & k = 2/c^2, \\ \text{literal } W^3 \text{ embedding}: & D = 2\rho P_{W^3}, & D_{ZZ}/\rho = 2c^2. \end{array} \tag{101}$$

The public chain must derive which map is meant and where the weak-mixing factor is normalized. The literal $2P_{W^3}$ tensor and the photon-safe $2c^2P_Z$ tensor are distinct cases: the first fails photon nullity, while the second passes it but changes the normalization.

G.6 Locked Diagnostics do not Isolate the Coefficient

The released arithmetic is

$$M_Z(k) = M_{Z,C1}\sqrt{1 + \Pi_{Z,C2} + \rho k}. \tag{102}$$

Monotonicity of the positive square root gives the exact released strict one-standard-deviation open interval

$$0.768434787688685769\ldots < k < 3.145092343548119919\ldots. \tag{103}$$

It is uncountable and its integer members are exactly $\{1, 2, 3\}$. Its endpoints have unit diagnostic distance and fail the released strict acceptance gate. The exact diagnostic central-value solution is $k = 1.956749882341106\ldots$, not two. Representative results are

Table 5: Parameter-free D_3 normalizations under the locked PR-III mass diagnostic.

k	$M_Z(k)$ (GeV)	absolute diagnostic distance (σ)
1	91.1859092277010	0.80513
$2\cos^2\theta_W = 1.53728847\ldots$	91.1868587299446	0.35299
2	91.1876764310439	0.03640
3	91.1894436001394	0.87790

M_W and G_F are independent of k in the released generator. Thus the locked diagnostics do not prove uniqueness and do not justify describing two as a best-fit selection. The PR-I quasar diagnostic is not connected to this PR-III coefficient by a released operator or propagation equation.

G.7 Public-source Ancestry and Exact Verdict

The released chain establishes the representation data and the stated candidate, but not the dynamical link between them. PR-III gives the reference-normalized determinant

$$\Delta\Gamma^{(1)}_{\mathrm{PR}}[B] = \frac{1}{2}\,\mathrm{STr}'_{\mathrm{PR}}\log[H_{\mathrm{PR}}[B]H_{\mathrm{PR}}[0]^{-1}], \tag{104}$$

including vector, ghost, scalar, and fermion blocks [6, 5]. Its electromagnetic charged-mode inventory is a Q_i^2 extraction for the photon kinetic term, not a derivation of the neutral D_3 mass Hessian. The manuscript then states Eq. 42; the public generator assigns

$$d_{3z} = \frac{2}{n_{\mathrm{gen}}}\epsilon_v^2 \tag{105}$$

and marks it as a candidate rather than an exact theorem. PR-III separately lists derivation of the full charged and neutral self-energy kernels as future work and explicitly distinguishes no-fit agreement from uniqueness. The missing noncircular edge is

$$\frac{1}{2}\,\mathrm{STr}'\log(H/H_0) \implies \text{complete order-}\rho\text{ neutral Hessian} \implies \rho\mathcal{E}(K) \tag{106}$$

with unit prefactor, no other same-order term, and $\mathcal{E}(P_3) = P_Z$. Until all H1–H5 ancestry is derived, the result is a complete conditional reduction theorem and a complete unconditional nonuniqueness theorem for the target-free public reduct plus the stated structural gates. Theorem G.2 supplies the formal axiom-by-axiom expansion proof. In particular:

- $K = 2P_3$ is proved exactly from the frozen representation;
- every $k \neq 2$ fails the canonical unit-prefactor identity;
- every tensor other than $2\rho P_Z$ fails a defining characterization defect, while the normalization defect remains explicitly underived physics; but
- even after imposing the three stipulated structural gates, the exact family $D(k) = \rho k P_Z$, and the released diagnostics permit a continuum of it.

Claiming unconditional P1–P7 uniqueness before deriving Eq. (106) would contradict the explicit counterfamily (97). The released-manuscript anchors are the PR-II generator-algebra, neutral-mixing, and charged-ladder sections, and the PR-III charged-mode-supertrace and neutral D_3 equations. Their stable labels and exact source-line ranges are listed in the accompanying public-source map.

H Physical-Z Admissibility and Exclusion Theorem

This appendix verifies that the physical-Z normalization used by the canonical D_3 closure is fixed and that apparent second-jet or effective-field-theory alternatives acquire freedom only by leaving the PR admissible class. No diagnostic residual is used.

H.1 Exact Physical Neutral Connection

PR-II supplies

$$\mathcal{A}_\mu^{\text{neutral}} = gW_\mu^3 T_3 + g' B_\mu \frac{Y}{2}, \qquad A = sW^3 + cB, \qquad Z = cW^3 - sB, \tag{107}$$

where $e = gs = g'c$, $Q = T_3 + Y/2$, and $c^2 + s^2 = 1$.

Proposition H.1 (Physical-Z generator and normalized charged-vector trace). *Define $t_Z = T_3 - s^2 Q$, $g_Z = g/c$, and let $q_Z = \mathrm{ad}_{g_Z t_Z}$ on $\mathrm{span}\{T^+, T^-\}$. Then*

$$\mathcal{A}_\mu^{\text{neutral}} = eA_\mu Q + g_Z Z_\mu t_Z, \qquad [g_Z t_Z, T^\pm] = \pm gc\, T^\pm. \tag{108}$$

Consequently the normalized physical charge operator $\widehat{q}_Z = (gc)^{-1} q_Z$ has spectrum $(+1, -1)$ and

$$\mathrm{Tr}_{\text{ch}}(\widehat{q}_Z^\dagger \widehat{q}_Z) = 2. \tag{109}$$

Proof. Inverting Eq. (107) gives $W^3 = sA + cZ$ and $B = cA - sZ$. The photon coefficient is $gsT_3 + g'cY/2 = eQ$, while the Z coefficient is $gcT_3 - g'sY/2 = (g/c)(T_3 - s^2 Q)$. On the weak-vector adjoint $Y = 0$, so the ladder commutators have eigenvalues $\pm gc$. Division by gc gives the normalized spectrum and trace. □

Thus the physical-Z map has exactly the unit-normalized charged-adjoint response assumed in the unique quadratic response Theorem. It supplies $F(\widehat{q}_Z) = 2$ without an adjustable mixing factor.

H.2 Second-jet Exclusion

For a normalized physical-Z background z, a reference-normalized determinant has second variation

$$\Gamma''(0) = \sum_i \eta_i \,\mathrm{Tr}'_i \left(G_i W_i - G_i V_i G_i V_i\right), \qquad V_i = H'_i(0), \quad W_i = H''_i(0). \tag{110}$$

Theorem H.2 (Independent second jets are outside the PR class). *Let one block be deformed by*

$$H_j^{(\lambda)}(z) = H_j^{(0)}(z) + \frac{\lambda\rho}{2\eta_j} z^2 H_j(0)^{1/2} X H_j(0)^{1/2}, \tag{111}$$

where X is positive, finite rank, commutes with the frozen charge action, and satisfies $\mathrm{Tr}' X = 1$. The zero and first jets are unchanged, but

$$\Gamma''_\lambda(0) - \Gamma''_0(0) = \lambda\rho. \tag{112}$$

For every independent $\lambda \neq 0$, the deformation introduces a new response datum and therefore violates inheritance, minimality, and the no-new-coefficient rules established in Section 2. It is not an alternative PR solution.

Proof. The deformation and its first derivative vanish at $z = 0$. Its second derivative changes only the $G_j W_j$ term in Eq. (110), by

$$\eta_j \,\mathrm{Tr}'\left[H_j(0)^{-1} \frac{\lambda\rho}{\eta_j} H_j(0)^{1/2} X H_j(0)^{1/2}\right] = \lambda\rho \,\mathrm{Tr}' X = \lambda\rho. \tag{113}$$

The free scalar λ is precisely an independent second-jet datum, which is forbidden by the completed minimal-response class. □

H.3 Effective-Operator Exclusion

Proposition H.3 (Independent neutral EFT operator is inadmissible)**.** *For the PR-II $Y = 1$ doublet, the gauge-invariant operator*

$$\mathcal{O}_{HD} = |\Phi^\dagger D_\mu \Phi|^2, \qquad \Delta\mathcal{L}_\lambda = \frac{2\lambda\rho}{v_{\rm EW}^2}\mathcal{O}_{HD} \tag{114}$$

produces at $\Phi_0 = (0, v_{\rm EW})^T/\sqrt{2}$

$$\Delta\mathcal{L}_\lambda\big|_{\Phi_0} = \frac{1}{2}\lambda\rho M_{Z,0}^2 Z_\mu Z^\mu, \qquad M_{Z,0}^2 = \frac{1}{4}g_Z^2 v_{\rm EW}^2. \tag{115}$$

It adds no charged-vector or photon mass, but its independent Wilson [9] coefficient λ and dimension-six operator violate the declared finite-tier operator inventory and no-new-parameter guideline. Hence it is a different theory, not a PR branch.

Proof. At the locked vacuum, $\Phi_0^\dagger D_\mu \Phi_0 = i v_{\rm EW}^2 g_Z Z_\mu/4$. Taking the modulus square gives the displayed neutral mass term. The operator carries an independent Wilson coefficient and changes the scalar/Goldstone interactions, so it is excluded by the complete PR class definition. □

Proposition H.4 (Physical-Z closure)**.** *Every fixed-kernel neutral response satisfying the PR inheritance, photon nullity, charged-sector preservation, canonical unit normalization, and minimal quadratic-response guidelines is operator-equivalent to*

$$\boxed{D_3 = \rho\,\mathrm{Tr}_{\rm ch}(\widehat{q}_Z^\dagger \widehat{q}_Z) P_Z = 2\rho P_Z.} \tag{116}$$

Every $k \neq 2$ has defect $\rho^2(k-2)^2$. Independent second jets and EFT operators have additional-data defects and lie outside $\mathfrak{A}_{\rm PR}^{\rm can}$. Therefore neither construction supplies a second admissible coefficient, equation, or solution branch.

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